

CCST5037: Mathematics: A Cultural Heritage
Examples of Central Themes for Group Project Essays

The following examples are by no means exhaustive, and you are encouraged to think of original theme ideas, but these may help you to get started and to form a general idea about what types of central themes for essays have proven successful in the past. The central themes appearing here are paraphrased versions of actual central themes appearing in past essays, but none of the phrases here actually appear in those essays.

- (1) In an essay relating to whether mathematics was discovered or invented, the following central theme was established: By investigating the interplay between the establishment of mathematics throughout history via the lenses of different periods of societal evolution, the mathematical geniuses who established this mathematics, and the ways in which theoretical development occurred, we find points and counterpoints supporting the claims that mathematics was discovered, ultimately concluding that certain aspects of mathematics were discovered, while others are the invention of the human mind.
- (2) In another essay about the impact of Calculus, the following central theme was established: The invention of Calculus by Newton has affected the development of nearly every scientific discipline since its inception. Prior to Newton's invention, the science of motion and change was still in its infancy, and its development has led to advances in engineering, medicine, business, meteorology, and even music. In this essay, we investigate how Newton brought us out of a static view of science and began a large movement in the scientific world.
- (3) In an essay based on the movie "Good Will Hunting", the following central theme was established: Mathematics helps to transform individuals' views of the world by teaching them to transform a collection of unstructured and imprecise facts into ordered and logical conclusions. The movie "Good Will Hunting" allows us to embark upon the metamorphosis that such mathematical development can bring about, even without proper formal education.
- (4) In an essay about the movie "Imitation Game", the following central theme was established: Like any tool, mathematics has the ability to serve as both a saviour and as a weapon in war. In this essay, we investigate the role that mathematics has played both in saving humanity from the disasters of war and in instigating atrocities that occurred within them.